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### ENHANCED HYBRID CLOCK

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#### Abstract

A subordinate clock includes a first servo loop, a second servo loop, and a third servo loop, each include a respective digitally controlled oscillator (DCO). The first DCO receives (i) a first error signal that is associated with a physical layer reference signal and (ii) an oscillating signal and outputs a physical layer clock signal. The second DCO receives (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal and outputs an intermediate clock signal. The third DCO receives (i) a third error signal associated with a grandmaster reference signal generated by a grandmaster clock and (ii) the intermediate clock signal and outputs a final clock signal with an average frequency that is generally equal to an average frequency of the grandmaster reference signal.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a U.S. national phase application, which is based on, and claims priority from, PCT application Serial No. PCT/US2023/033998, filed Sep. 28, 2023, which is based on, and claims priority from, which claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/411,393, filed on Sep. 29, 2022, which are hereby incorporated by reference herein in their entirety.

### TECHNICAL FIELD

[0002] Embodiments of the present invention relate generally to timing systems, and more specifically, to methods and apparatus for achieving hybrid clock operation within timing systems.

### BACKGROUND

[0003] Packet-based timing methods are becoming essential for delivering timing over packet-switched networks, often referred to as the cloud. New systems and methods for enabling clocks to utilize multiple references are needed.

### SUMMARY

[0004] According to a first implementation of the present disclosure, a subordinate clock comprises a first servo loop, a second servo loop, and a third servo loop. The first servo loop includes a first digitally controlled oscillator (DCO) that is configured to receive (i) a first error signal that is associated with a physical layer reference signal and (ii) an oscillating signal, and to output a physical layer clock signal. The second servo loop includes a second DCO that is configured to receive (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal, and to output an intermediate clock signal. The third servo loop includes a third DCO that is configured to receive (i) a third error signal associated with a grandmaster reference signal generated by a grandmaster clock and (ii) the intermediate clock signal, and to output a final clock signal. An average frequency of the final clock signal is generally equal to an average frequency of the grandmaster reference signal.

[0005] In some aspects of the first implementation, each of the servo loops has a phase-locked loop structure.

[0006] In some aspects of the first implementation, the first servo loop includes a first comparator configured to receive (i) the physical layer reference signal and (ii) the physical layer clock signal from the first DCO, and to output the first error signal. In some aspects of the first implementation, the first error signal is associated with a difference between the physical layer reference signal and the physical layer clock signal.

[0007] In some aspects of the first implementation, the second servo loop includes a second comparator configured to receive (i) the physical layer clock signal and (ii) the intermediate clock signal from the second DCO, and to output the second error signal.

[0008] In some aspects of the first implementation, the second error signal is associated with a difference between the physical layer clock signal and the intermediate clock signal.

[0009] In some aspects of the first implementation, the third servo loop includes a third comparator configured to receive (i) the grandmaster reference signal and (ii) the final clock signal from the third DCO, and to output the third error signal.

[0010] In some aspects of the first implementation, the third error signal is indicative of an offset between a time of the grandmaster clock and a time of the subordinate clock.

[0011] In some aspects of the first implementation, the first servo loop has a frequency bandwidth and (i) attenuates noise associated with the physical layer reference signal that is above the frequency bandwidth of the first servo loop, and (ii) attenuates noise associated with the oscillating

signal that is below the frequency bandwidth of the first servo loop.

[0012] In some aspects of the first implementation, the frequency bandwidth of the first servo loop is between about 1 Hz and about 10 Hz, between about 3 Hz and about 10 Hz, or about 3 Hz.

[0013] In some aspects of the first implementation, the second servo loop has a frequency bandwidth and (i) attenuates the noise associated with the physical layer reference signal that is above the frequency bandwidth of the second servo loop, and (ii) attenuates the noise associated with the oscillating signal that is below the frequency bandwidth of the second servo loop.

[0014] In some aspects of the first implementation, the frequency bandwidth of the second servo loop is between about 0.01 Hz and about 0.1 Hz, about 0.1 Hz, or about 0.01 Hz.

[0015] In some aspects of the first implementation, the third servo loop has a frequency bandwidth and (i) attenuates noise associated with the grandmaster reference signal that is above the frequency bandwidth of the third servo loop, and (ii) attenuates the noise associated with the physical layer reference signal that is below the frequency bandwidth of the third servo loop.

[0016] In some aspects of the first implementation, the frequency bandwidth of the third servo loop is between about 0.05 Hz and about 0.1 Hz or about 0.1 Hz.

[0017] In some aspects of the first implementation, the physical layer reference signal is a signal generated by a Synchronous Optical Networking (SONET) transmission link, a signal generated by a Synchronous Digital Hierarchy (SDH) transmission link, a signal generated by a Synchronous Ethernet (SyncE) transmission link, or a timing input from a Building Integrated Timing Supply (BITS) system, and wherein the grandmaster reference signal is Precision Time Protocol (PTP) timing input signal.

[0018] In some aspects of the first implementation, the third servo loop operates to drive an offset between a time of the subordinate clock and a time of the grandmaster clock to zero or to a prescribed constant value.

[0019] According to a second implementation of the present disclosure, a timing system comprises a grandmaster clock, a physical layer, and a subordinate clock. The grandmaster clock is configured to generate a grandmaster reference signal associated with a time of the grandmaster clock. The physical layer is configured to generate a physical layer reference signal associated with a frequency of the physical layer. The subordinate clock includes a first servo loop, a second servo loop, and a third servo loop. The first servo loop includes a first digitally controlled oscillator (DCO) that is configured to receive (i) a first error signal that is associated with the physical layer reference signal and (ii) an oscillating signal, and to output a physical layer clock signal. The second servo loop includes a second DCO that is configured to receive (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal, and to output an intermediate clock signal. The third servo loop includes a third DCO that is configured to receive (i) a third error signal associated with a grandmaster reference signal generated by the grandmaster clock and (ii) the intermediate clock signal, and to output a final clock signal. An average frequency of the final clock signal is generally equal to an average frequency of the grandmaster reference signal.

[0020] In some aspects of the second implementation, the second servo loop attenuates noise associated with the physical layer reference signal that is above a frequency bandwidth of the second servo loop, the frequency bandwidth of the second servo loop being between about 0.01 Hz and about 0.1 Hz.

[0021] In some aspects of the second implementation, the first servo loop has a frequency bandwidth of between about 1 Hz, and about 10 Hz, and (i) attenuates noise associated with the physical layer reference signal that is above the frequency bandwidth of the first servo loop, and (ii) attenuates noise associated with the oscillating signal that is below the frequency bandwidth of the first servo loop.

[0022] In some aspects of the second implementation, the second servo loop has a frequency bandwidth of between about 0.01 Hz and about 0.1 Hz, and (i) attenuates the noise associated with

the physical layer reference signal that is above the frequency bandwidth of the second servo loop, and (ii) attenuates the noise associated with the oscillating signal that is below the frequency bandwidth of the second servo loop.

[0023] In some aspects of the second implementation, the third servo loop has a frequency bandwidth of between about 0.05 Hz and about 0.1 Hz, and (i) attenuates noise associated with the grandmaster reference signal that is above the frequency bandwidth of the third servo loop, and (ii) attenuates the noise associated with the physical layer reference signal that is below the frequency bandwidth of the third servo loop.

[0024] In some aspects of the second implementation, the third servo loop operates to drive an offset between a time of the subordinate clock and a time of the grandmaster clock to zero or to a prescribed constant value.

[0025] The above summary is not intended to represent each implementation or every aspect of the present disclosure. Additional features and benefits of the present disclosure are apparent from the detailed description and figures set forth below.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a conceptual diagram that depicts functional blocks of a PTP Grandmaster clock, according to aspects of the present disclosure.

[0027] FIG. 2 identifies the time-stamps and events associated with the timing signal suitable for transferring timing from a packet-based master clock to a packet-based subordinate clock, according to aspects of the present disclosure.

[0028] FIG. 3 is a diagram that depicts blocks of a subordinate clock, according to aspects of the present disclosure.

[0029] FIG. 4 is a diagram of a two-loop subordinate clock, according to aspects of the present disclosure.

[0030] FIG. 5A is a mathematical depiction of a servo loop shows the mathematical operations effectively performed on signals within the servo loop, according to aspects of the present disclosure.

[0031] FIG. 5B shows the frequency response characteristics of the servo loop of FIG. 5A, according to aspects of the present disclosure.

[0032] FIG. 6 is a diagram of a virtual loop system showing the filtering results of the subordinate clock of FIG. 4, according to aspects of the present disclosure.

[0033] FIG. 7A shows the frequency response characteristics of a first servo loop of the subordinate clock of FIG. 4, according to aspects of the present disclosure. FIG. 7B shows the frequency response characteristics of a second servo loop of the subordinate clock of FIG. 4, according to aspects of the present disclosure.

[0034] FIG. 7A shows the combined frequency response characteristics of the first servo loop and the second servo loop of the subordinate clock of FIG. 4, according to aspects of the present disclosure.

[0035] FIG. 8 is a diagram of a three-loop subordinate clock, according to aspects of the present disclosure.

[0036] FIG. 9 is a diagram of a virtual loop system showing the filtering results of the subordinate clock of FIG. 8, according to aspects of the present disclosure.

[0037] FIG. 10A shows a time error of the two-loop clock of FIG. 4 compared to a mask, according to aspects of the present disclosure.

[0038] FIG. 10B shows a time error of the three-loop clock of FIG. 8 compared to the mask, according to aspects of the present disclosure.

[0039] While the present disclosure is susceptible to various modifications and alternative forms, specific implementations and embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that it is not intended to limit the present disclosure to the particular forms disclosed, but on the contrary, the present disclosure is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

#### DETAILED DESCRIPTION

[0040] Precision Timing Protocol (PTP) is often used as a standard for delivering timing information (time/phase/frequency) from a grandmaster (GM) clock (sometimes referred to as a grandmaster clock or a grandmaster) to subordinate clocks in end application-specific equipment, such as wireless base stations providing mobile telephony services that require precise timing and the backhaul method of choice is Ethernet.

[0041] The grandmaster clock provides timing information over the packet-switched network to the subordinate clocks by exchanging packets with embedded time-stamps related to the time-of-arrival and time-of-departure of the timing packets. The subordinate clock utilizes this information to align its time (and frequency of its time-clock) with the grandmaster clock. The grandmaster clock is provided an external reference to serve as the basis for time and often frequency as well. Most commonly this reference is derived from a Global Navigation Satellite System (GNSS) such as the global positioning system (GPS) System that in turn is controlled by the US Department of Defense and its timing controlled very precisely and linked to the US Naval Observatory. Time alignment to the GPS clock is, for all practical purposes, equivalent to time alignment to UTC.

[0042] In many cases other timing references are available to the subordinate clock, such as frequency references derived from physical layer signals such as Synchronous Ethernet (SyncE) transmission links, Synchronous Optical Network (SONET) transmission links, or Synchronous Digital Hierarchy (SDH) transmission links; or timing inputs from an Office Building Integrated Timing Supply (BITS) system that are traceable to a primary reference source (PRS) or primary reference clock (PRC). In such instances, it is advantageous to implement a hybrid subordinate clock with two servo loops (e.g., phase-locked loops), where one servo loop is used to accept the frequency reference and the other servo loop utilizes the physical layer clock as the driver, and disciplines this clock to drive the Time-of-Day counter, that in turn is the source for the time-stamps struck for the time-of-arrival and time-of-departure of the packets involved in the packet-based synchronization of the subordinate clock to its master.

[0043] The physical layer clock component of the subordinate clock is not unconstrained. For example, if the subordinate clock is embedded in a device that is part of a physical layer synchronization network such SyncE, SONET, or SDH, then the physical layer clock must satisfy the requirements associated with that architecture. The packet-layer clock must accept this timing signal even if it has not been optimized.

[0044] FIG. 1 is a simplified block diagram representing a grandmaster (GM) clock **100** (based on PTP). The GM **100** receives a time input **105**, which can be in the form of a 1PPS (one-pulse-per-second) signal together with a message channel indicating the precise time-of-day at the significant event (rising edge or falling edge, depending on convention) of the 1PPS signal. The time and frequency generation is done by a packet-based clock **112** that can be part of a greater packet engine **110**. The GM **100** includes a local frequency output **106**, and local time outputs **107** and **108**. The GM **100**, acting as the master, receives the precise time-stamps **114** of incoming and outgoing packets and inserts the appropriate time-stamps into outgoing packets. Time-stamping is performed by a packet-processing block **116** that uses a local clock signal **150** from an oscillator module **125** that is part of a physical layer clock **120**. In some implementations, the physical layer clock **120** is synchronized using the 1PPS signal as a 1 Hz reference frequency input. Packets enter the GM **100** via packet input **130A** and exit the GM **100** via packet output **130B**, which may be formed using traditional packet I/O methods, such as Ethernet. An output providing the physical

layer clock 135 is common. The GM 100 provides a timing reference to down-stream subordinate clocks. The manner in which the subordinate clocks align themselves with the master is illustrated in FIG. 2. Packet exchanges between a master 202 (e.g., the GM 100) and a subordinate 204 provide measurements of the transit delay between the two. The particular protocol employed determines the method whereby the measurements (“time-stamps”) are communicated between the two entities. This protocol can be, for example, PTP or network time protocol (NTP). Both are supported by packet networks in a transparent fashion.

[0045] The sequence of events and important items of information associated with an exchange of packets between master and subordinate includes events A, B, C, and D. Event A includes a first packet being transmitted by the GM 100 with a time of departure of. G-Event B includes the first packet arriving at the subordinate 204 which measures the time of arrival of the packet as  $T_{sub.2}$ . If the time of the subordinate 204 is offset from the time of the master 202 by an offset  $a$ , the actual time of the arrival of the packet at the subordinate 204 is  $t_{sub.2} = T_{sub.2} + E$ . Event C includes a second packet being transmitted by the subordinate 204 that notes the time of departure as  $T_{sub.3}$ . Given the offset  $a$  between the time of the master 202 and the subordinate 204, the actual time of departure of the second packet from the subordinate 204 is  $t_{sub.3} = T_{sub.3} + a$ . Event D includes the second packet arriving at the master 202, which measures the time of arrival as  $t_{sub.4}$ .

[0046] In PTP, the first packet going from the master 202 to the subordinate 204 (denoted by the arrow from A to B) is called a “Sync Message” and carries the time-stamp  $T_1$  whose value is G-Providing the time of departure of a packet in the packet itself is difficult to do precisely. Consequently, in PTP, there is a notion of a “Follow-on Message”. The Sync Message contains an approximate value of  $T_1$  whereas the precise value of  $T_1$ , e.g.,  $G$ , is carried by a Follow-on Message which, as the name implies, is transmitted shortly after the Sync Message. The Follow-on Message is not depicted in FIG. 2 since it is not timed but just carries information (the precise value of  $T_1$ ). The time-stamp struck at the subordinate 204 upon arrival of the Sync Message is called  $T_2$  and the value, according to the time clock in the subordinate clock is  $T_2$ . The second packet going from the subordinate 204 to the master 202 (denoted by the arrow from C to D) is called a “Delay Request” and has a time of departure of  $T_3$ , which has a value of  $T_3$  according to the time clock of the subordinate 204. The master 202 time-stamps the time of arrival of the Delay Request at the master 202, and this time-stamp is called  $T_4$  and has the value  $t_{sub.4}$ . The master 202 then sends this value back to the subordinate 204 in a Delay Response message. This is not a timed message and is not explicitly shown in FIG. 2. The subordinate 204 thus has the times of arrival and times of departure of all Sync Messages and Delay Request messages, often referred to as the four-tuple ( $T_1, T_2, T_3, T_4$ ).

[0047] Such a two-way exchange of packets can provide information suitable for allowing the subordinate 204 to align in time with the master 202. If the exchange of information is only one-way, from the master 202 to the subordinate 204 (referred to as the forward direction), the subordinate 204 can still align its clock (frequency) with the master 202 since the first packet contains the time-of-departure from the master ( $Z_q$ ) and the subordinate 204 measures the time-of-arrival ( $T_2$ ). This alignment process is referred to as syntonization. One-way methods, where the time-stamped packets flow from the subordinate 204 to the master 202 can be employed provided the mechanism is available for the subordinate 204 to obtain the results of the master 202 measuring time of arrival at the master ( $t_{sub.4}$ ).

[0048] There are four measured values that can be communicated between the master 202 and the subordinate 204, namely,  $t_{sub.1}(T_{sub.2}, T_{sub.3}, t_{sub.4}$ . Such a two-way exchange involves one packet sent in each direction. These packets do not have to be consecutive, as long as the time stamp information is communicated appropriately. In some instances, the rate at which packets are transmitted in the two directions can be different. Denoting by  $A_{sub.MS}$  and  $A_{sub.SM}$  as the transit delays from the master 202 to the subordinate 204 and from the subordinate 204 to the master 202 respectively, the following equations (referred to as Eq. 1) can be established:

$t_q = T_{sub.2} + E + ASM$  (for a packet from the master **202** to the subordinate **204**); and  $t_{sub.4} = T_{sub.3} + \epsilon + A_{sub.SM}$  (for a packet from the subordinate **204** to the master **202**).

[0049] In an actual time-transfer situation there are two equations with three unknowns, so it is common to assume reciprocity of transit delay between the two devices, thereby reducing the number of unknowns to two, and therefore computing  $s$ , the subordinate time offset from master according to the following equation (referred to as Eq. 2):  $s = \{ \text{circumflex over } ( \quad ) \} 4 \cdot \sup. + t_{i.\sup.} T_3 + \sup. T_2 > (t_4 - T_{sub.3}) - (T_{sub.2} - t_1) / 2$ .

[0050] Because of the fundamental statistical behavior of packet networks, the transit delays are not fixed and can vary from packet to packet. To counter this packet delay variation, as well as to account for any drift in the clock oscillator of the subordinate **204**, the estimates of clock offset are made routinely, and it is well known that the mitigation of the deleterious effects of packet delay variation and oscillator drift is improved by using more frequent exchanges of timing packets. Simple subordinate clocks develop their estimate of time offset from master based on (Eq. 2) and use this information to discipline their time-clock in both frequency and time/phase.

[0051] Hybrid subordinate clocks utilize the availability of a frequency reference embedded in the transmission fabric of the telecommunication network. Thus, when the transmission is based on Time-Division-Multiplexing (TDM), such as in SONET/SDH (Synchronous Optical NETwork/Synchronous Digital Hierarchy) or Ethernet based on the principles of Synchronous Ethernet (aka “SyncE”) a physical layer timing reference (frequency) is available to the device.

[0052] A hybrid subordinate clock **300** is depicted in FIG. 3. The subordinate clock **300** includes a PTP layer **310** and a physical layer **320**, which both have servo loop structures. The PTP layer **310** receives a PTP time input **330** (or another packet-based timing reference) and based on the principles described above aligns its time clock **312** with the master. The time clock **312** can generate an 1PPS output **351** (or other timing output) and/or provide a PTP master port and provide a PTP time output **350** to down-stream subordinates. The physical layer (frequency) clock **322** labeled EEC/SEC (for “Ethernet Equipment Clock/Synchronous Equipment Clock) accepts the physical layer frequency input reference **340** and uses PLL means to generate a frequency-synchronized physical layer clock signal **360** that is used in the physical layer transmission (e.g., SyncE). This physical layer clock signal **360** is used as the oscillating signal to drive the PTP layer **310** (e.g., the packet layer) to assist in the packet layer synchronization. This frequency assist provided by the physical layer **320** to the PTP layer **310** is the basis for the term “hybrid” or “assisted” clock.

[0053] FIG. 4 shows the components of a hybrid subordinate clock **400**, which is generally the same as or similar to the subordinate clock **300** of FIG. 3. The PHY loop **410** represents the PLL (phase locked loop) that provides the physical layer clock signal **416**. This is achieved by using a digitally controlled oscillator (DCO) **415** for adjusting the frequency of the clock signal originating from the local oscillator **430**. Methods and techniques for implementing this DCO function are well known in the field of PLLs. The control signal that is provided to the DCO **415** to implement the frequency adjustment is developed by comparing the generated physical layer clock signal **416** to the input physical layer reference signal **411** using a phase comparator **412**. The phase comparator **412** generates the difference, often referred to as “phase error signal” or “error signal” that is processed using servo control techniques (such as proportional control and/or integral control) in the control block **413**. In modern devices this processing is done digitally with reasonably large word-lengths in the arithmetic. The block labeled Q **414** represents the restriction of word-length, if necessary, to align with the input word-length requirements of the DCO **415**, and can be referred to as a word-length restriction block. The selector **440** permits the choice of either the oscillator **430** itself or the physical layer clock signal **416** as the oscillating signal **427** that is input to a DCO **425** of a PTP loop **420**. The PTP loop **420** is also a closed servo loop and is depicted with similar functional blocks as the PHY loop **410**. The PTP loop **420** generates a PTP clock signal **426** by adjusting the frequency of the incoming oscillating signal **427** using the DCO **425**. The PTP clock

signal **426** is used to drive a counter driven, where the counter value represents time. The rate of the PTP clock signal **426** determines the granularity of time that is representable. For example, if the rate is 125 MHz, each clock cycle of the time-stamping clock increments the counter by one unit which is considered, in time units, as 8 ns; a 100 MHz time-stamping clock is associated with a time-granularity of 10 ns, and so on. The higher the rate of the PTP clock signal **426** the more precision is available in the time representation.

[0054] The phase comparator **422** represents the development of the “time error” for the PTP loop **420**. The PTP reference signal **421** is, in effect, the stream of time-stamps (T1, T2, T3, T4) where the values  $t_{sub.4}$  and  $t_{sub.4}$  are based on the time-clock of the master and  $T_{sub.2}$  and  $T_{sup.3}$  based on the time-clock in the DCO **425**. The time-error is equivalent to the time offset generated as in Eq. 2. The similarity in pictorial representation of **422** and **412** is solely to indicate that they both generate an error signal based on the reference input and the clock signal of that loop (**426** for time and **416** for frequency) suitable for driving the servo control loop that develops the control value for the DCO (**425** or **415**). In general, the servo loop is used to drive this time error to zero. However, in some implementations, the servo loop is used to drive this time error to a prescribed constant value by adding the prescribed constant value (which may be positive or negative) to the calculated offset.

[0055] In summary, FIG. 4 depicts a hybrid subordinate clock **400** that has two servo loops, one for time (PTP loop **420**) and one for frequency (PHY loop **410**) and it is possible to use the physical layer clock signal **416** as the oscillating mechanism for the PTP loop **420** to achieve the hybrid operation.

[0056] The underlying mathematical and signal processing principles of a general phase-locked-loop are depicted in the virtual loop **500** of a PLL in FIG. 5A. When we are in a locked condition, the important signals are the fluctuations, i.e., noise signals (also referred to as “jitter” or “wander”), and the signal processing conditions these fluctuations to minimize the impact in the output. The signal  $\{e_i(n)\}$  is the noise in the input reference;  $\{e_o(n)\}$  is the noise in the oscillator signal; and  $\{e_o(n)\}$  is the noise in the output signal (output clock). This representation of signals is standard in discrete time signal processing, namely,  $\{x(n)\}$  is a signal named “x” that is sampled at a given sampling rate and the notion of  $x(n)$  is the  $n$ -th sample, taken at time instant  $(nT_s)$  where  $T_s$  is the sampling interval that is the reciprocal of the sampling rate  $f_s$ .

[0057] The virtual loop **500** depicted in FIG. 5 assumes that the output rate is  $N$  times the rate of the input and hence needs to be divided by  $N$  at block **525** in order to implement the phase difference operation at block **505**. The digital filter block **510** represents the proportional and integral control filtering, and acts as a low-pass filter (also referred to as an “LPF”) on the noise present in the input. It is well known that phase is the integral of frequency and so the model for an oscillator is a digital integrator **520** as shown. The noise in the oscillator is additive to the DCO and is shown as a superposition by the summer **530**.

[0058] In FIG. 5A, there are two noise signals that are input into the virtual loop **500**, noise signal **540** that is present in the oscillator signal and noise signal **550** that is present in the reference input. The effective signal processing that is achieved on noise signal **540** and noise signal **550** is depicted in terms of the frequency response characteristic **560** in FIG. 5B, which measures the attenuation  $(|H(f)|)$  versus the frequency of the noise measured in Hz.

[0059] Specifically, the noise signal **540** in the oscillator signal experiences a high-pass filter characteristic in its path to the output as shown in the attenuation line **562**, whereas the noise signal **550** in the reference input experiences a low-pass filter characteristic in its path to the output as shown in attenuation line **564**. Generally speaking, the noise in the input reference tends to be composed of higher frequency components and can be attenuated by the low-pass characteristic and the noise in the oscillator signal tends to be composed of lower frequency components and can be attenuated by the high-pass characteristic. That is why the servo loop is effective in reducing noise.



[0060] The high-pass and low-pass characteristics are not independent but have the same 3-dB cut-off frequency. It is common to associate the low-pass behavior with a “time-constant” that is roughly inversely proportional to the cut-off frequency of the low-pass characteristic. Thus, a large time-constant is associated with a narrow-band filter. A narrow-band filter will be effective in attenuating noise in the reference input, but the trade-off is that since the oscillator sees the high-pass version, more oscillator noise will be transferred to the input. It is well known that for very narrow-band loops it is necessary to have very stable oscillators such as oven-controlled crystal oscillators (OCXOs) or temperature-compensated crystal oscillators (TCXOs).

[0061] Legacy Synchronous Equipment Clocks (SONET/SDH) and Ethernet Equipment Clocks (SyncE) specified a servo loop bandwidth of between 3 Hz and 10 Hz for the physical layer clock; modem clocks use Enhanced SyncE and require a bandwidth of between 1 Hz and 3 Hz. In contrast, the PTP layer requires a much narrower bandwidth loop. Certain clocks are for packet-based clocks with the nominal packet rate of 16 packets/second (for Sync Messages and Delay Request messages) and a servo loop bandwidth of between 0.05 Hz and 0.1 Hz.

[0062] The efficacy of hybrid clocks arises from the use of the SyncE reference as a source of frequency accuracy and of frequency stability. In normal operation, the SyncE reference can be traced back to a Primary Reference Clock (PRC) for its accuracy. The relatively wide-band (between 1 Hz and 10 Hz) servo loops imply that when there is a chain of SyncE physical layer clocks, the oscillators in the clocks do not add much noise and hence the hybrid (subordinate) clock can expect, in normal operation, an accurate, stable, reference input. There may be some high frequency jitter components associated with clock recovery of the physical layer reference but that tends to be removable by the servo loop (cutoff frequency between 1 Hz and 10 Hz). This makes the physical layer clock a very good candidate for providing the oscillating signal to the PTP layer clock.

[0063] FIG. 6 shows a virtual loop system **600** that models the filter effects of the subordinate clock **400** when the selector directs the physical layer clock signal **416** to the DCO **425** instead of the output of the oscillator **430**. Virtual loop **610** represents the PHY loop **410**, and is modeled as having a loop bandwidth of 3 Hz. The physical layer reference signal **411** experiences a 3 Hz low-pass filter **614** between input and the physical layer clock signal **416**. The oscillator **430** signal experiences a 3 Hz high-pass filter **612** between the oscillator and the physical layer clock signal **416**. Virtual loop **620** represents the PTP loop **420**, and is modeled as having a loop bandwidth of 0.1 Hz. The PTP reference signal **421** experiences a 0.1 Hz low-pass filter **624** between input and the PTP clock signal **426**. The physical layer clock signal **416** experiences a 0.1 Hz high-pass filter **622** between the input and the PTP clock signal **426**.

[0064] FIGS. 7A-7C show the response characteristics of the subordinate clock **400**. Response characteristic **710** shows the attenuation ( $|H(f)|$ ) in the PHY loop **410** versus the frequency of the noise measured in Hz. Attenuation line **712** shows the attenuation between the oscillator **430** and the output of the physical layer clock signal **416** due to the 3 Hz high-pass filter effect of the PHY loop **410**. Attenuation line **714** shows the attenuation between the input of the physical layer reference signal **411** and the output of the physical layer clock signal **416** due to the 3 Hz low-pass filter effect of the PHY loop **410**.

[0065] Response characteristic **720** shows the attenuation ( $|H(f)|$ ) in the PTP loop **420** versus the frequency of the noise measured in Hz. Attenuation line **722** shows the attenuation between the oscillator **430** and the output of the PTP clock signal **426** (if SyncE is not enabled) or between the output of the physical layer clock signal **416** and the output of the PTP clock signal **426** (if SyncE is enabled), due to the 0.1 Hz high-pass filter effect of the PTP loop **420**. Attenuation line **724** shows the attenuation between the input of the PTP reference signal **421** and the output of the PTP clock signal **426** due to the 0.1 Hz low-pass filter effect of the PTP loop **420**.

[0066] Response characteristic **730** shows the combined attenuation ( $|H(f)|$ ) in the PHY loop **410** and the PTP loop **420** versus the frequency of the noise measured in Hz. Attenuation line **732**

shows the attenuation between the oscillator **430** and the output of the PTP clock signal **426** due to the combination of the 3 Hz high-pass filter effect of the PHY loop **410** and the 0.1 Hz high-pass filter effect of the PTP loop **420**. Attenuation line **734** shows the attenuation between the input of the physical layer reference signal **411** and the output of the PTP clock signal **426**. The response characteristic **730** identifies the impact of the dual loop arrangement of the noise (jitter) present in the physical layer reference signal **411** on the PTP clock signal **426** as a band-pass filter with passband range from 0.1 Hz to 3 Hz.

[0067] Thus, the dual-loop hybrid clock has numerous advantages. First, the PTP layer is provided an accurate frequency because the SyncE network is timed by a Primary Reference Clock (PRC) that is more accurate than any standalone oscillator. This helps, for example, for “holdover” when there is an interruption in the packet-layer timing reference and the clock has to continue its time-clock based on the provided oscillating signal **427** from the oscillator **430** or the physical layer clock signal **416**. Second, the high-pass effect of the PHY loop **410** has a cut-off frequency of the order of 1 Hz to 10 Hz depending on the SyncE clock specification. This permits the utilization of a less expensive oscillator. However, one disadvantage of the dual-loop hybrid clock is that low-frequency wander components present in the physical layer reference signal **411** can be transferred to the PTP clock signal **426** due to the band-pass characteristics resulting from the combination of the PHY loop **410** and the PTP loop **420**, shown in the response characteristic **730**. This can be substantial during periods of rearrangement in the SyncE network, which are not uncommon.

[0068] The enhanced method described here retains the primary advantage of the dual-loop hybrid clock, namely the frequency accuracy of the physical layer reference signal **411**. By recognizing that high-performance oscillators are becoming less expensive, the enhanced method utilizes the stability of these high-performance oscillators to overcome the disadvantage of the dual-loop method, namely the instabilities in the physical layer reference input that can be caused by, for example, rearrangements in the SyncE network.

[0069] FIG. **8** illustrates a three-loop subordinate clock **800** that includes a first servo loop **810**, a second servo loop **820**, a third servo loop **830**, and an oscillator **840**. Each of these servo loops **810-830** can have a phase-locked loop structure (e.g., the output of each loop is fed back into the input, and the input and output are compared to further adjust the output to match the input). The first servo loop **810** is generally the same as the PHY loop **410** of the subordinate clock **400** and can also be referred to as the PHY loop **810**. The first servo loop **810** includes a comparator **812** (which may be a phase comparator that is the same as or similar to the phase comparator **412**, and determines a phase difference between its two inputs), a control block **813** (the same as or similar to the control block **413**), a word-length restriction block **814** (the same as or similar to the word-length restriction block **414**), and a digitally controlled oscillator (DCO) **815** (the same as or similar to the DCO **415**).

[0070] Similarly, the third servo loop **830** is generally the same as the PTP loop **420** of the subordinate clock **400** and can also be referred to as the PTP layer loop **830**. The third servo loop **830** includes a comparator **832** (which may be a phase comparator that is the same as or similar to the phase comparator **422**, and determines a phase difference between its two inputs), a control block **833** (the same as or similar to the control block **423**), a word-length restriction block **834** (the same as or similar to the word-length restriction block **424**), and a digitally controlled oscillator (DCO) **835** (the same as or similar to the DCO **425**), the output of which is used to generate the local time of the subordinate clock **800** by driving a time-stamping clock, such as the time-stamping clock described herein.

[0071] The second servo loop **820** is connected between the first servo loop **810** and the third servo loop **830**, and includes a comparator **822** (which may be a phase comparator that is the same as or similar to the phase comparator **412** and/or the phase comparator **422**, and determines the phase difference between its two inputs), a control block **823** (the same as or similar to the control block **413**), a word-length restriction block **824** (the same as or similar to the word-length restriction

block **414**), and a digitally controlled oscillator (DCO) **825** (the same as or similar to the DCO **415**).

[0072] As used herein, the individual components of each of the three servo loops **810**, **820**, **830** can also be referred to using ordinal numbers. Thus, the subordinate clock **800** can be said to include a first comparator **812**, a second comparator **822**, and a third comparator **832**; a first control block **813**, a second control block **823**, and a third control block **833**; a first word-length restriction block **814**, a second word-length restriction block **824**, and a third word-length restriction block **834**; and a first DCO **815**, a second DCO **825**, and a third DCO **835**.

[0073] The first servo loop **810** is electrically connected in the same fashion as the PHY loop **410**. The first comparator **812** receives a physical layer reference signal **811** and a physical layer clock signal **816**, which is the output of the first DCO **815**. The physical layer reference signal **811** is generally the same as the physical layer reference signal **411** and can be any signal with a stable frequency from a system that the subordinate clock **800** is being used with. For example, the physical layer reference signal **811** can be a signal that is generated by a Synchronous Optical Networking (SONET) transmission link, a signal generated by a Synchronous Digital Hierarchy (SDH) transmission link, a signal generated by a Synchronous Ethernet (SyncE) transmission link, a timing input from a Building Integrated Timing Supply (BITS) system, other signals, or any combinations thereof.

[0074] The first comparator **812** generates a first error signal that is fed into the first control block **813**. The first error signal is associated with the difference (e.g., the phase difference) between the physical layer reference signal **811** and the physical layer clock signal **816** that is output by the first DCO **815**. The first control block **813** processes the first error signal and passes it to the first word-length restriction block **814**, which performs any necessary restriction of the first error signal as required by the first DCO **815**. The first error signal is then fed into the first DCO **815**, which also receives an oscillating signal **841** that is generated by the oscillator **840**. The first DCO **815** generates the physical layer clock signal **816**, which as noted is fed back into the first comparator **812**. The physical layer clock signal **816** is the same as or similar to the physical layer clock signal **416** and is a stable clock signal based on the stable frequency of the physical layer reference signal **811**. By feeding the physical layer clock signal **816** back into the first comparator **812**, the first DCO **815** adjusts the frequency of the oscillating signal **841** to match the frequency of the physical layer reference signal **811**, which is output as the physical layer clock signal **816**.

[0075] The physical layer clock signal **816** is also fed into the second comparator **822** of the second servo loop **820**, along with an intermediate clock signal **826**, which is the output of the second DCO **825**. The second servo loop **820** operates in generally the same fashion as the first servo loop **810**. The second comparator **822** generates a second error signal that is associated with the difference (e.g., the phase difference) between the physical layer clock signal **816** and the intermediate clock signal **826**, which is then processed by the control block **823** and restricted (if necessary) by the word-length restriction block **824**. The second error signal is then fed into the second DCO **825**, which also receives the oscillating signal **841** that is generated by the oscillator **840**. The second DCO **825** generates the intermediate clock signal **826**, which as noted is fed back into the second comparator **822**. By feeding the intermediate clock signal **826** back into the second comparator **822**, the second DCO **825** adjusts the frequency of the oscillating signal **841** to match the frequency of the physical layer clock signal **816**, which is output as the intermediate clock signal **826**. Thus, the intermediate clock signal **826** generally has the same frequency as the physical layer clock signal **816**, which is based on the stable frequency of the physical layer reference signal **811**.

[0076] The third servo loop **830** is electrically connected in the same fashion as the PTP loop **420**. The third comparator **832** receives a grandmaster reference signal **831** and a final clock signal **836**, which is the output of the third DCO **835**. The grandmaster reference signal **831** is generally the same as the PTP reference signal **421**, and can be generated by a grandmaster clock (also referred

to herein as “the grandmaster”), such as the grandmaster clock **100** of FIG. **1**. In some implementations, the grandmaster reference signal **831** is a Precision Timing Protocol (PTP) timing input as discussed herein. The grandmaster reference signal **831** is generally associated with the time of the grandmaster.

[0077] The third servo loop **830** operates in generally the same fashion as the first servo loop **810** and the second servo loop **820** (and also the same fashion as the PTP loop **420**). The third comparator **832** generates a third error signal that is associated with the difference (e.g., the phase difference) between the grandmaster reference signal **831** and the final clock signal **836**, which is then processed by the control block **833** and restricted (if necessary) by the wordlength restriction block **834**. The phase difference between the grandmaster reference signal **831** and the final clock signal **836** is equivalent to the time difference between the grandmaster time (represented by the grandmaster reference signal **831**) and the local time of the subordinate clock **800** (represented by the final clock signal **836**). The third error signal is then fed into the third DCO **835**, which also receives the intermediate clock signal **826** that is generated by the second DCO **825**. The third DCO **835** generates the final clock signal **836**, which is associated with the time offset between the grandmaster time and the local time of the subordinate clock **800**. By then feeding the final clock signal **836** back into the third comparator **832**, the third DCO **835** operates to adjust the frequency of the intermediate clock signal **826** to match the frequency of the grandmaster reference signal **831** such that the time offset between the grandmaster and the subordinate clock **800** (represented by the final clock signal **836**) is driven to zero, or to a prescribed constant value by including slight variation as needed. Thus, after the third servo loop **830** has converged, the final clock signal **836** has an average frequency that is generally equal to the average frequency as the grandmaster reference signal **831** and is generated using the stable frequency of the physical layer reference signal **811** via the intermediate clock signal **826** and the physical layer clock signal **816**. Thus, the subordinate clock **800** can be used to perform packed-based synchronization. In some implementations, the average frequency of the final clock signal **836** is exactly equal to the average frequency of the grandmaster reference signal **831**. In these implementations, while the frequency of the final clock signal **836** at a specific point in time may vary slightly relative the frequency of the grandmaster reference signal **831**, the average frequency of the final clock signal **836** be equal to the average frequency of the grandmaster reference signal **831**. However, in other implementations, the average frequency itself of the final clock signal **836** may vary (temporarily or permanently) relative to the average frequency of the grandmaster reference signal **831**. As used to describe the average frequencies of the final clock signal **836** and the grandmaster reference signal **831**, the term “generally equal” can include exactly equal, within  $\pm 1\%$ , within  $\pm 2\%$ , within  $\pm 3\%$ , within  $\pm 4\%$ , within  $\pm 5\%$ , within  $\pm 6\%$ , within  $\pm 7\%$ , within  $\pm 8\%$ , within  $\pm 9\%$ , within  $\pm 10\%$ , within  $\pm 1\%-5\%$ , within  $\pm 5\%-10\%$ , within  $\pm 1\%-10\%$ , or any combinations thereof.

[0078] To drive the difference between the grandmaster and the subordinate clock **800** to a non-zero constant value, an additional term is added to the error signal generated by the third comparator **832**. This term may be positive or negative depending on whether the subordinate clock **800** is needed to be set ahead of the grandmaster or behind the grandmaster. This term can be added directly after the third error signal is output from the third comparator **832**, after the third error signal has been processed by the control block **833**, after the third error signal has been processed by the word-length restriction block **834**, or any time before the third error signal is input into the third DCO **835**.

[0079] FIG. **9** illustrates a frequency response model **900** of the subordinate clock **800**. As shown, the first servo loop **810** is modeled as a first virtual loop **910** that includes a first high-pass filter **912** (HPFi) and a first low-pass filter **914** (LPFi). The second servo loop **820** is modeled as a second virtual loop **920** that includes a second high-pass filter **922** (HPF2) and a second low-pass filter **924** (LPF2). The third servo loop **830** is modeled as a third virtual loop **930** that includes a third high-pass filter **932** (HPF3) and a third low-pass filter **934** (LPF3).

[0080] As shown, in its path from the physical layer to the third servo loop **830**, the physical layer reference signal **811** passes through the first low-pass filter **914** in the first virtual loop **910**, the second low-pass filter **924** in the second virtual loop **920**, and the third high-pass filter **932** of the third virtual loop **930**. The physical layer reference signal **811** thus passes through two low-pass filters instead of the single low-pass filter **614** that the physical layer reference signal **411** passes through in the virtual loop system **600** of the subordinate clock **400**. In its path from the oscillator **840** to the third servo loop **830**, the oscillating signal **841** passes through the first high-pass filter **912** and the second high-pass filter **922**. Finally, the grandmaster reference signal **831** passes through the third low-pass filter **934**. Thus, the physical layer reference signal **811** effectively passes through a band-pass filter formed from the first low-pass filter **914**, the second low-pass filter **924**, and the third high-pass filter **932**. The oscillating signal **841** passes through a dual-stage filter formed from the first high-pass filter **912** and the second high-pass filter **922**. By designing the servo loops **810-830** to have certain frequency bandwidths, noise associated with the various signal can be attenuated and/or removed. For example, noise that was otherwise not attenuated and/or removed from the physical layer reference signal **411** is attenuated and/or removed from the physical layer reference signal **811**.

[0081] The first servo loop **810** attenuates noise associated with the physical layer reference signal **811** that is above the frequency bandwidth of the first servo loop **810**, and attenuates noise associated with the oscillating signal **841** that is below the frequency bandwidth of the first servo loop **810**. In some implementations, the frequency bandwidth of the first servo loop **810** is between about 1 Hz and about 10 Hz, between about 3 Hz and about 10 Hz, or about 3 Hz. The bandwidth restriction on the first servo loop **810** is imposed by certain standards that specify the behavior of transmission systems such as Synchronous Ethernet (e.g., ITU-T Rec. G.8262, ITU-T Rec. G.8262.1, and others).

[0082] The second servo loop **820** attenuates noise associated with the physical layer reference signal **811** that is above the frequency bandwidth of the second servo loop **820**, and attenuates noise associated with the oscillating signal **841** that is below the frequency bandwidth of the second servo loop **820**. In some implementations, the frequency bandwidth of the second servo loop **820** is between about 0.01 Hz and about 0.1 Hz, about 0.1 Hz, or about 0.01 Hz.

[0083] The third servo loop **830** attenuates noise associated with the grandmaster reference signal **831** that is above the frequency bandwidth of the third servo loop **830**, and attenuates noise associated with the physical layer reference signal **811** that is below the frequency bandwidth of the third servo loop **830**. In some implementations, the frequency bandwidth of the third servo loop **830** is between about 0.05 Hz and about 0.1 Hz, or about 0.1 Hz.

[0084] In some implementations, the frequency bandwidth of the first servo loop **810** is about 3 Hz, the frequency bandwidth of the second servo loop **820** is about 0.1 Hz, and the frequency bandwidth of the third servo loop **830** is about 0.1 Hz. In these implementations, the physical layer reference signal **811** is effectively filtered by a bandpass filter where the upper end of the passband is at 0.1 Hz (from the second low-pass filter **924** in the second virtual loop **920**), and the lower end of the passband is also at 0.1 Hz (from the third high-pass filter **932** in the third virtual loop **930**). If the second servo loop **820** did not exist, then the upper end of the passband would be at 3 Hz (from the first low-pass filter **914**), and noise associated with the physical layer reference signal between 0.1 Hz and 3 Hz would not be attenuated and/or removed. Thus, by adding the second servo loop **820** that has a bandwidth that is less than the third servo loop **830**, a vast majority of the noise associated with the physical layer reference signal **811** is attenuated and/or removed.

[0085] FIG. **10A** shows a plot of the time error **1020** (measured in nanoseconds) versus elapsed time (measured in seconds) of the two-loop subordinate clock **400** compared to a mask **1010**, which is effectively a maximum time error allowed when there is a rapid change in the physical layer (under the conditions of the test performed to generate the plot). FIG. **10B** shows a plot of the time error **1030** versus elapsed time of the three-loop subordinate clock **800** compared to the same

mask **1010**. As shown, the time error **1020** of the two-loop subordinate clock **400** violates the mask **1010**, whereas the time error **1030** of the three-loop subordinate clock **800** does not violate the mask **1010**.

[0086] The three-loop subordinate clock **800** can be used in conjunction with a grandmaster clock and a physical layer clock to form a timing system that is relatively immune to any wander or noise in the physical layer clock. As discussed, the grandmaster clock can be configured to generate a PTP timing input signal that forms the grandmaster reference signal **831**. The physical clock can be part of any physical layer that may be used, including a SONET transmission link, an SDH transmission link, a SyncE transmission link, a BITS system, and others, which in many instances require the bandwidth of the servo loop that generates the physical layer clock to be between 1.0 Hz and 10.0 Hz.

[0087] Various aspects of the present disclosure can be implemented using any suitable combination of hardware and/or software. For example, the subordinate clock **800** can be implemented using PLL-specific integrated circuits that contain various components of the servo loops **810-830**. One or more processors (e.g., microcontrollers) can additionally or alternatively be used to perform various functions of the subordinate clock **800** and/or to process any of the various signals that are input into the subordinate clock **800**, that are output from the subordinate clock **800**, or are propagating within the subordinate clock **800**.

#### ALTERNATIVE IMPLEMENTATIONS

[0088] Alternative Implementation 1. A subordinate clock comprising: a first servo loop containing a first digitally controlled oscillator (DCO), the first DCO configured to receive (i) a first error signal that is associated with a physical layer reference signal and (ii) an oscillating signal, and to output a physical layer clock signal; a second servo loop containing a second DCO, the second DCO configured to receive (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal, and to output an intermediate clock signal; and a third servo loop containing a third DCO, the third DCO configured to receive (i) a third error signal associated with a grandmaster reference signal generated by a grandmaster clock and (ii) the intermediate clock signal, and to output a final clock signal, wherein an average frequency of the final clock signal is equal to an average frequency of the grandmaster reference signal.

[0089] Alternative Implementation 2. The subordinate clock of claim **1**, wherein each of the servo loops has a phase-locked loop structure.

[0090] Alternative Implementation 3. The subordinate clock of claim **1**, wherein the first servo loop includes a first comparator configured to receive (i) the physical layer reference signal and (ii) the physical layer clock signal from the first DCO, and to output the first error signal.

[0091] Alternative Implementation 4. The subordinate clock of claim **3**, wherein the first error signal is associated with a difference between the physical layer reference signal and the physical layer clock signal.

[0092] Alternative Implementation 5. The subordinate clock of claim **3**, wherein the second servo loop includes a second comparator configured to receive (i) the physical layer clock signal and (ii) the intermediate clock signal from the second DCO, and to output the second error signal.

[0093] Alternative Implementation 6. The subordinate clock of claim **5**, wherein the second error signal is associated with a difference between the physical layer clock signal and the intermediate clock signal.

[0094] Alternative Implementation 7. The subordinate clock of claim **5**, wherein the third servo loop includes a third comparator configured to receive (i) the grandmaster reference signal and (ii) the final clock signal from the third DCO, and to output the third error signal.

[0095] Alternative Implementation 8. The subordinate clock of claim **7**, wherein the third error signal is indicative of an offset between a time of the grandmaster clock and a time of the subordinate clock.

[0096] Alternative Implementation 9. The subordinate clock of claim **1**, wherein the first servo loop

has a frequency bandwidth and (i) attenuates noise associated with the physical layer reference signal that is above the frequency bandwidth of the first servo loop, and (ii) attenuates noise associated with the oscillating signal that is below the frequency bandwidth of the first servo loop. [0097] Alternative Implementation 10. The subordinate clock of claim **9**, wherein the frequency bandwidth of the first servo loop is between about 1 Hz and about 10 Hz, between about 3 Hz and about 10 Hz, or about 3 Hz. Alternative Implementation 11. The subordinate clock of claim **9**, wherein the second servo loop has a frequency bandwidth and (i) attenuates the noise associated with the physical layer reference signal that is above the frequency bandwidth of the second servo loop, and (ii) attenuates the noise associated with the oscillating signal that is below the frequency bandwidth of the second servo loop.

[0098] Alternative Implementation 12. The subordinate clock of claim **11**, wherein the frequency bandwidth of the second servo loop is between about 0.01 Hz and about 0.1 Hz, about 0.1 Hz, or about 0.01 Hz.

[0099] Alternative Implementation 13. The subordinate clock of claim **11**, wherein the third servo loop has a frequency bandwidth and (i) attenuates noise associated with the grandmaster reference signal that is above the frequency bandwidth of the third servo loop, and (ii) attenuates the noise associated with the physical layer reference signal that is below the frequency bandwidth of the third servo loop.

[0100] Alternative Implementation 14. The subordinate clock of claim **9**, wherein the frequency bandwidth of the third servo loop is between about 0.05 Hz and about 0.1 Hz or about 0.1 Hz.

[0101] Alternative Implementation 15. The subordinate clock of claim **1**, wherein the physical layer reference signal is a signal generated by a Synchronous Optical Networking (SONET) transmission link, a signal generated by a Synchronous Digital Hierarchy (SDH) transmission link, a signal generated by a Synchronous Ethernet (SyncE) transmission link, or a timing input from a Building Integrated Timing Supply (BITS) system, and wherein the grandmaster reference signal is Precision Time Protocol (PTP) timing input signal.

[0102] Alternative Implementation 16. The subordinate clock of claim **1**, wherein the third servo loop operates to drive an offset between a time of the subordinate clock and a time of the grandmaster clock to zero or to a prescribed constant value.

[0103] Alternative Implementation 17. A timing system comprising: a grandmaster clock configured to generate a grandmaster reference signal associated with a time of the grandmaster clock; a physical layer configured to generate a physical layer reference signal associated with a frequency of the physical layer; and a subordinate clock including: a first servo loop containing a first digitally controlled oscillator (DCO), the first DCO configured to receive (i) a first error signal that is associated with the physical layer reference signal and (ii) an oscillating signal, and to output a physical layer clock signal; a second servo loop containing a second DCO, the second DCO configured to receive (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal, and to output an intermediate clock signal; and a third servo loop containing a third DCO, the third DCO configured to receive (i) a third error signal associated with the grandmaster reference signal from the grandmaster clock and (ii) the intermediate clock signal, and to output a final clock signal, wherein an average frequency of the final clock signal is equal to an average frequency of the grandmaster reference signal.

[0104] Alternative Implementation 18. The timing system of claim **17**, wherein the second servo loop attenuates noise associated with the physical layer reference signal that is above a frequency bandwidth of the second servo loop, the frequency bandwidth of the second servo loop being between about 0.01 Hz and about 0.1 Hz.

[0105] Alternative Implementation 19. The timing system of claim **17**, wherein the first servo loop has a frequency bandwidth of between about 1 Hz, and about 10 Hz, and (i) attenuates noise associated with the physical layer reference signal that is above the frequency bandwidth of the first servo loop, and (ii) attenuates noise associated with the oscillating signal that is below the

frequency bandwidth of the first servo loop.

[0106] Alternative Implementation 20. The timing system of claim **17**, wherein the second servo loop has a frequency bandwidth of between about 0.01 Hz and about 0.1 Hz, and (i) attenuates the noise associated with the physical layer reference signal that is above the frequency bandwidth of the second servo loop, and (ii) attenuates the noise associated with the oscillating signal that is below the frequency bandwidth of the second servo loop.

[0107] Alternative Implementation 21. The timing system of claim **17**, wherein the third servo loop has a frequency bandwidth of between about 0.05 Hz and about 0.1 Hz, and (i) attenuates noise associated with the grandmaster reference signal that is above the frequency bandwidth of the third servo loop, and (ii) attenuates the noise associated with the physical layer reference signal that is below the frequency bandwidth of the third servo loop.

[0108] Alternative Implementation 22. The timing system of claim **17**, wherein the third servo loop operates to drive an offset between a time of the subordinate clock and a time of the grandmaster clock to zero or to a prescribed constant value.

[0109] One or more elements or aspects or steps, or any portion(s) thereof, from one or more of any of the Alternative Implementations and/or claims herein can be combined with one or more elements or aspects or steps, or any portion(s) thereof, from one or more of any of the other Alternative Implementations and/or claims herein or combinations thereof, to form one or more additional implementations and/or claims of the present disclosure.

[0110] While the present disclosure has been described with reference to one or more particular embodiments or implementations, those skilled in the art will recognize that many changes **1** may be made thereto without departing from the spirit and scope of the present disclosure. Each of these implementations and obvious variations thereof is contemplated as falling within the spirit and scope of the present disclosure. It is also contemplated that additional implementations according to aspects of the present disclosure may combine any number of features from any of the implementations described herein.

## Claims

**1.** A subordinate clock comprising: a first servo loop containing a first digitally controlled oscillator (DCO), the first DCO configured to receive (i) a first error signal that is associated with a physical layer reference signal and (ii) an oscillating signal, and to output a physical layer clock signal; a second servo loop containing a second DCO, the second DCO configured to receive (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal, and to output an intermediate clock signal; and a third servo loop containing a third DCO, the third DCO configured to receive (i) a third error signal associated with a grandmaster reference signal generated by a grandmaster clock and (ii) the intermediate clock signal, and to output a final clock signal, wherein an average frequency of the final clock signal is generally equal to an average frequency of the grandmaster reference signal.

**2.** The subordinate clock of claim 1, wherein each of the servo loops has a phase-locked loop structure.

**3.** The subordinate clock of claim 1, wherein the first servo loop includes a first comparator configured to receive (i) the physical layer reference signal and (ii) the physical layer clock signal from the first DCO, and to output the first error signal.

**4.** The subordinate clock of claim 3, wherein the first error signal is associated with a difference between the physical layer reference signal and the physical layer clock signal.

**5.** The subordinate clock of claim 3, wherein the second servo loop includes a second comparator configured to receive (i) the physical layer clock signal and (ii) the intermediate clock signal from the second DCO, and to output the second error signal.

**6.** The subordinate clock of claim 5, wherein the second error signal is associated with a difference



between the physical layer clock signal and the intermediate clock signal.

**7.** The subordinate clock of claim 5, wherein the third servo loop includes a third comparator configured to receive (i) the grandmaster reference signal and (ii) the final clock signal from the third DCO, and to output the third error signal.

**8.** The subordinate clock of claim 7, wherein the third error signal is indicative of an offset between a time of the grandmaster clock and a time of the subordinate clock.

**9.** The subordinate clock of claim 1, wherein the first servo loop has a frequency bandwidth and (i) attenuates noise associated with the physical layer reference signal that is above the frequency bandwidth of the first servo loop, and (ii) attenuates noise associated with the oscillating signal that is below the frequency bandwidth of the first servo loop.

**10.** The subordinate clock of claim 9, wherein the frequency bandwidth of the first servo loop is between about 1 Hz and about 10 Hz, between about 3 Hz and about 10 Hz, or about 3 Hz.

**11.** The subordinate clock of claim 9, wherein the second servo loop has a frequency bandwidth and (i) attenuates the noise associated with the physical layer reference signal that is above the frequency bandwidth of the second servo loop, and (ii) attenuates the noise associated with the oscillating signal that is below the frequency bandwidth of the second servo loop.

**12.** The subordinate clock of claim 11, wherein the frequency bandwidth of the second servo loop is between about 0.01 Hz and about 0.1 Hz, about 0.1 Hz, or about 0.01 Hz.

**13.** The subordinate clock of claim 11, wherein the third servo loop has a frequency bandwidth and (i) attenuates noise associated with the grandmaster reference signal that is above the frequency bandwidth of the third servo loop, and (ii) attenuates the noise associated with the physical layer reference signal that is below the frequency bandwidth of the third servo loop.

**14.** The subordinate clock of claim 9, wherein the frequency bandwidth of the third servo loop is between about 0.05 Hz and about 0.1 Hz or about 0.1 Hz.

**15.** The subordinate clock of claim 1, wherein the physical layer reference signal is a signal generated by a Synchronous Optical Networking (SONET) transmission link, a signal generated by a Synchronous Digital Hierarchy (SDH) transmission link, a signal generated by a Synchronous Ethernet (SyncE) transmission link, or a timing input from a Building Integrated Timing Supply (BITS) system, and wherein the grandmaster reference signal is Precision Time Protocol (PTP) timing input signal.

**16.** The subordinate clock of claim 1, wherein the third servo loop operates to drive an offset between a time of the subordinate clock and a time of the grandmaster clock to zero or to a prescribed constant value.

**17.** A timing system comprising: a grandmaster clock configured to generate a grandmaster reference signal associated with a time of the grandmaster clock; a physical layer configured to generate a physical layer reference signal associated with a frequency of the physical layer; and a subordinate clock including: a first servo loop containing a first digitally controlled oscillator (DCO), the first DCO configured to receive (i) a first error signal that is associated with the physical layer reference signal and (ii) an oscillating signal, and to output a physical layer clock signal; a second servo loop containing a second DCO, the second DCO configured to receive (i) a second error signal associated with the physical layer clock signal and (ii) the oscillating signal, and to output an intermediate clock signal; and a third servo loop containing a third DCO, the third DCO configured to receive (i) a third error signal associated with the grandmaster reference signal from the grandmaster clock and (ii) the intermediate clock signal, and to output a final clock signal, wherein an average frequency of the final clock signal is generally equal to an average frequency of the grandmaster reference signal.

**18.** The timing system of claim 17, wherein the second servo loop attenuates noise associated with the physical layer reference signal that is above a frequency bandwidth of the second servo loop, the frequency bandwidth of the second servo loop being between about 0.01 Hz and about 0.1 Hz.

**19.** The timing system of claim 17, wherein the first servo loop has a frequency bandwidth of

between about 1 Hz, and about 10 Hz, and (i) attenuates noise associated with the physical layer reference signal that is above the frequency bandwidth of the first servo loop, and (ii) attenuates noise associated with the oscillating signal that is below the frequency bandwidth of the first servo loop.

**20.** The timing system of claim 17, wherein the second servo loop has a frequency bandwidth of between about 0.01 Hz and about 0.1 Hz, and (i) attenuates the noise associated with the physical layer reference signal that is above the frequency bandwidth of the second servo loop, and (ii) attenuates the noise associated with the oscillating signal that is below the frequency bandwidth of the second servo loop.

**21.** The timing system of claim 17, wherein the third servo loop has a frequency bandwidth of between about 0.05 Hz and about 0.1 Hz, and (i) attenuates noise associated with the grandmaster reference signal that is above the frequency bandwidth of the third servo loop, and (ii) attenuates the noise associated with the physical layer reference signal that is below the frequency bandwidth of the third servo loop.

**22.** The timing system of claim 17, wherein the third servo loop operates to drive an offset between a time of the subordinate clock and a time of the grandmaster clock to zero or to a prescribed constant value.

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